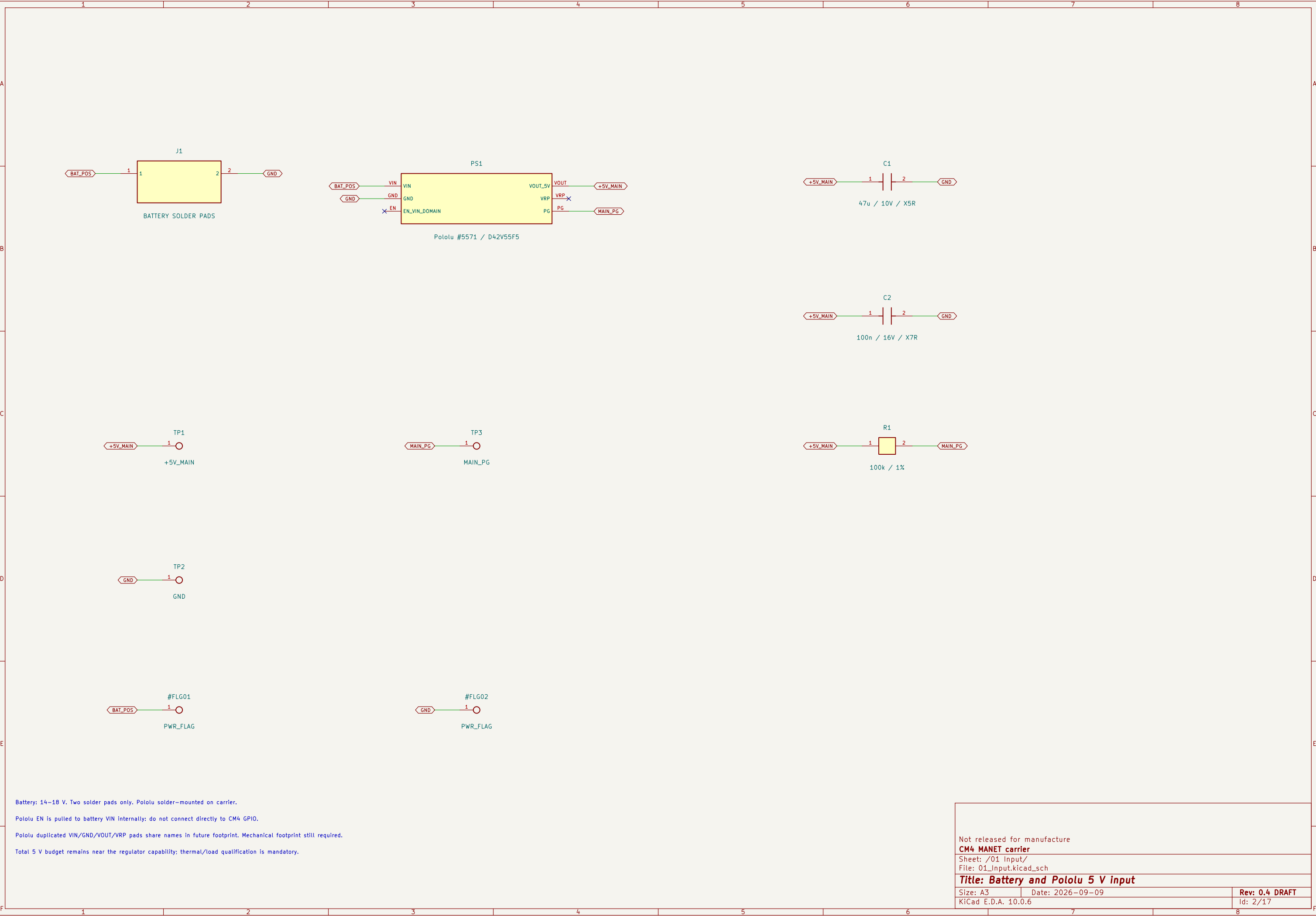
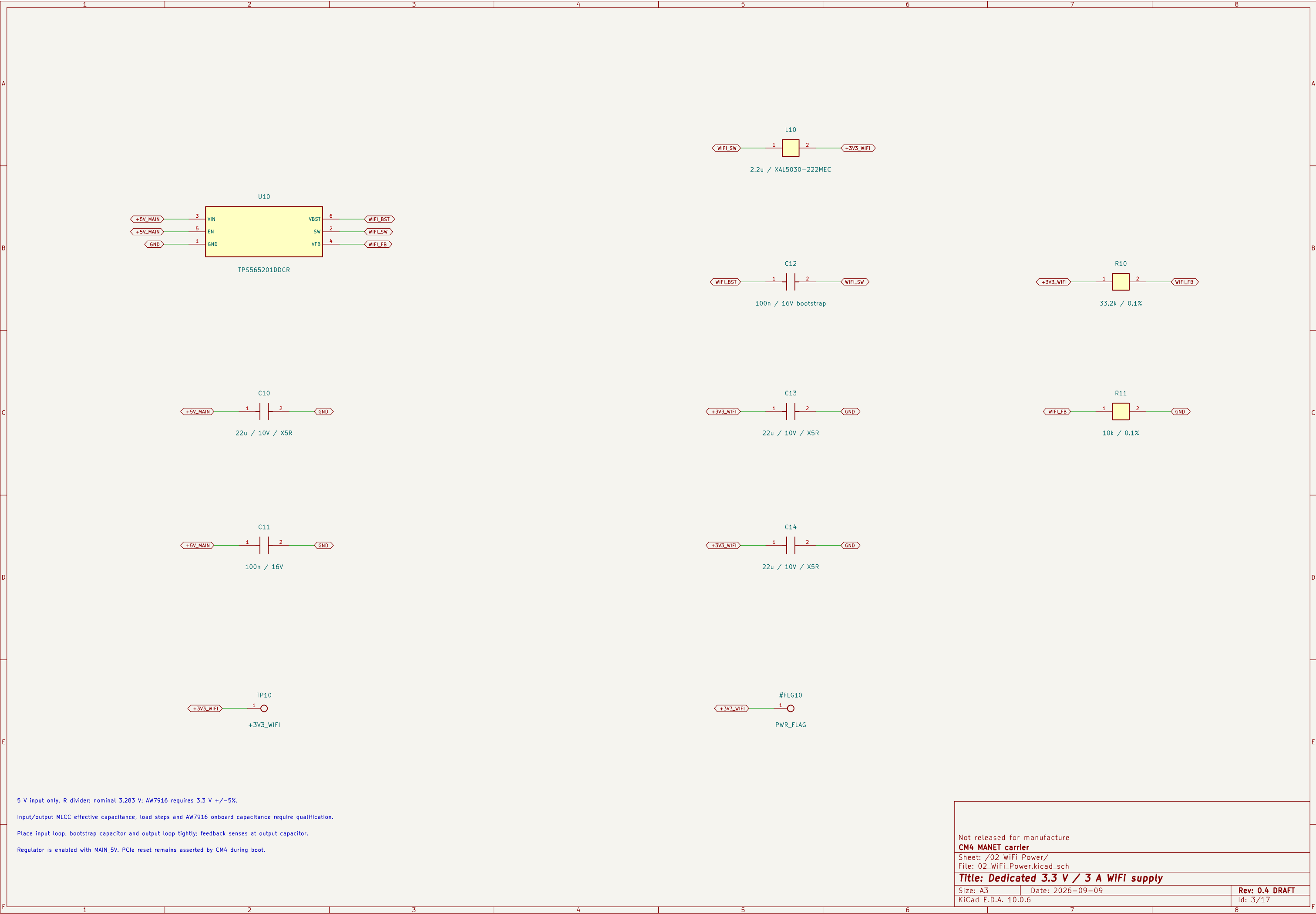


	1	2	3	4	5	6	7	8
A	01 Input Page 2 / 01 Input File: 01_Input.kicad_sch		02 WiFi Power Page 3 / 02 WiFi Power File: 02_WiFi_Power.kicad_sch		03 Aux Power Page 4 / 03 Aux Power File: 03_Aux_Power.kicad_sch		04 CM4 Power GPIO Page 5 / 04 CM4 Power GPIO File: 04_CM4_Power_GPIO.kicad_sch	
B	05 CM4 HighSpeed Page 6 / 05 CM4 HighSpeed File: 05_CM4_HighSpeed.kicad_sch		06 Service Reset Page 7 / 06 Service Reset File: 06_Service_Reset.kicad_sch		07 MicroSD Page 8 / 07 MicroSD File: 07_MicroSD.kicad_sch		08 USB Hub Page 9 / 08 USB Hub File: 08_USB_Hub.kicad_sch	
C	09 USB Clock Decoupling Page 10 / 09 USB Clock Decoupling File: 09_USB_Clock_Decoupling.kicad_sch		10 USB A Page 11 / 10 USB A File: 10_USB_A.kicad_sch		11 USB B Page 12 / 11 USB B File: 11_USB_B.kicad_sch		12 HaLow Power Page 13 / 12 HaLow Power File: 12_HaLow_Power.kicad_sch	
D	13 HaLow RF Page 14 / 13 HaLow RF File: 13_HaLow_RF.kicad_sch		14 WiFi PCIe Page 15 / 14 WiFi PCIe File: 14_WiFi_PcLe.kicad_sch		15 Ethernet Cable Page 16 / 15 Ethernet Cable File: 15_Ethernet_Cable.kicad_sch		16 Panel Interfaces Page 17 / 16 Panel Interfaces File: 16_Panel_Interfaces.kicad_sch	
E	DRAFT: complete functional sheet set; manufacturing release blocked by open component/footprint checks. CM4 top; AW7916 bottom; carrier target 85 x 56mm. 4-layer PCB not routed yet. All sheet interfaces use global labels. PanelInterfaces sheet is off-carrier wiring reference. External interfaces: 2 x USB2 500mA and 1 x GbE via JST GH prototype harnesses, maximum 10cm. See LEESMIJ.md and ONTWERPCONTROLE.md for validation, cable pinouts and unresolved items.							
F							USB and Ethernet ports moved to enclosure via cables <=10cm Sheet: / File: CM4_MANET.kicad_sch Title: CM4 MANET carrier – integrated schematic Size: A3Date: 2026-09-09Rev: 0.4 DRAFT KiCad E.D.A. 10.0.6Id: 1/17	





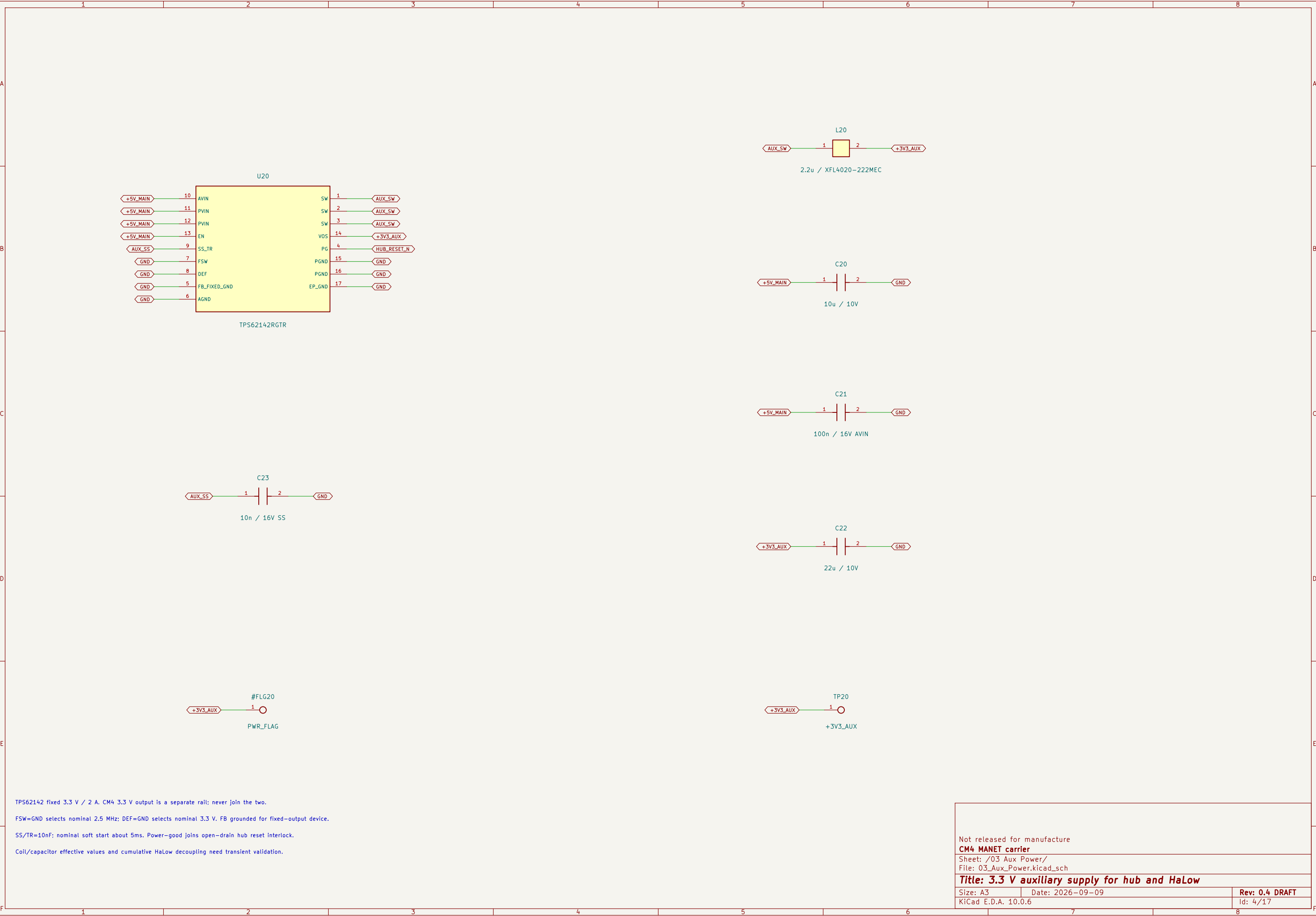
5 V input only. R divider: nominal 3.283 V; AW7916 requires 3.3 V +/-5%.

Input/output MLCC effective capacitance, load steps and AW7916 onboard capacitance require qualification.

Place input loop, bootstrap capacitor and output loop tightly; feedback senses at output capacitor.

Regulator is enabled with MAIN_5V. PCIe reset remains asserted by CM4 during boot.

Not released for manufacture		
CM4 MANET carrier		
Sheet: /02_WiFi_Power/		
File: 02_WiFi_Power.kicad_sch		
Title: Dedicated 3.3 V / 3 A WiFi supply		
Size: A3	Date: 2026-09-09	Rev: 0.4 DRAFT
KiCad E.D.A. 10.0.6		Id: 3/17



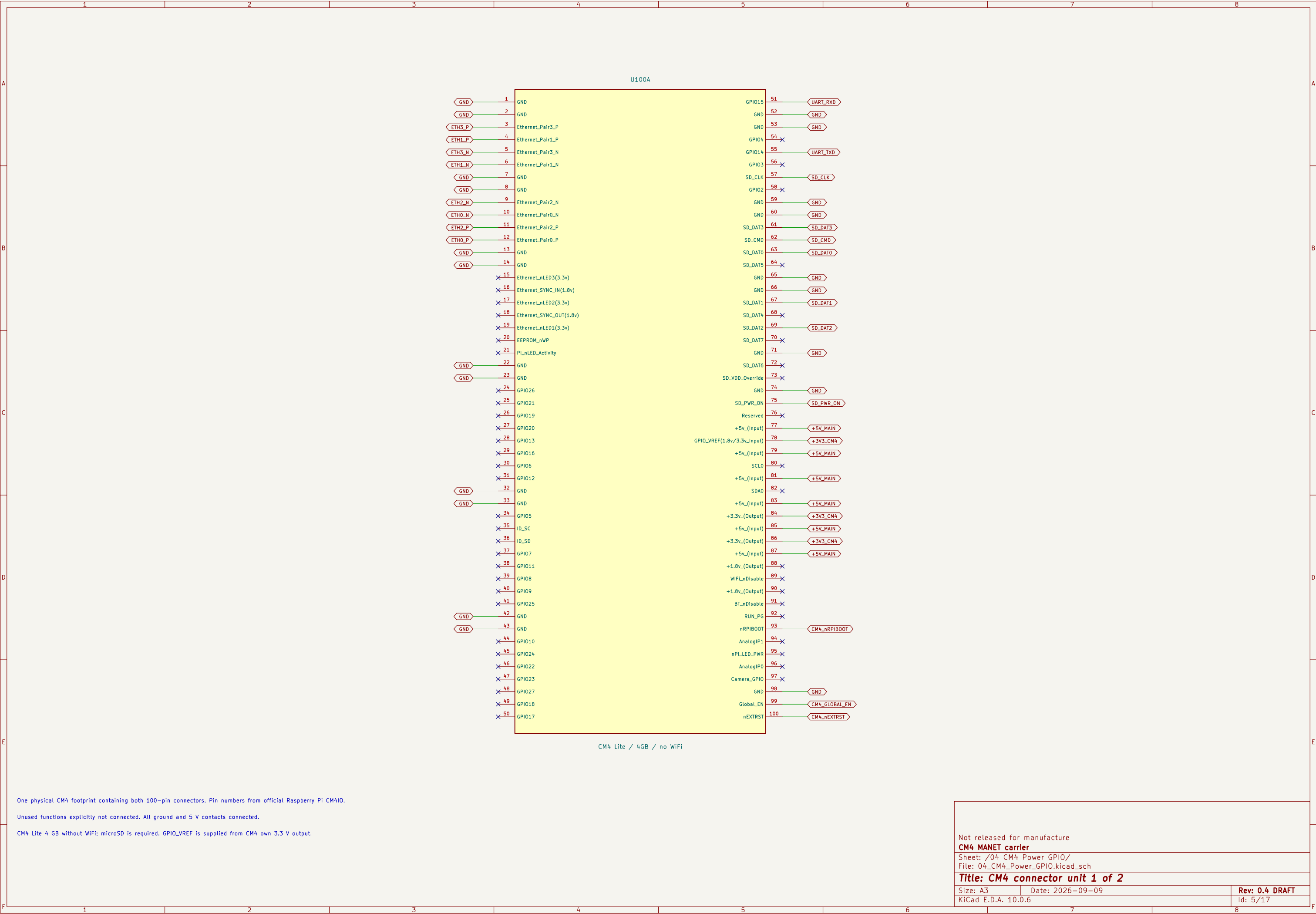
TPS62142 fixed 3.3 V / 2 A. CM4 3.3 V output is a separate rail; never join the two.

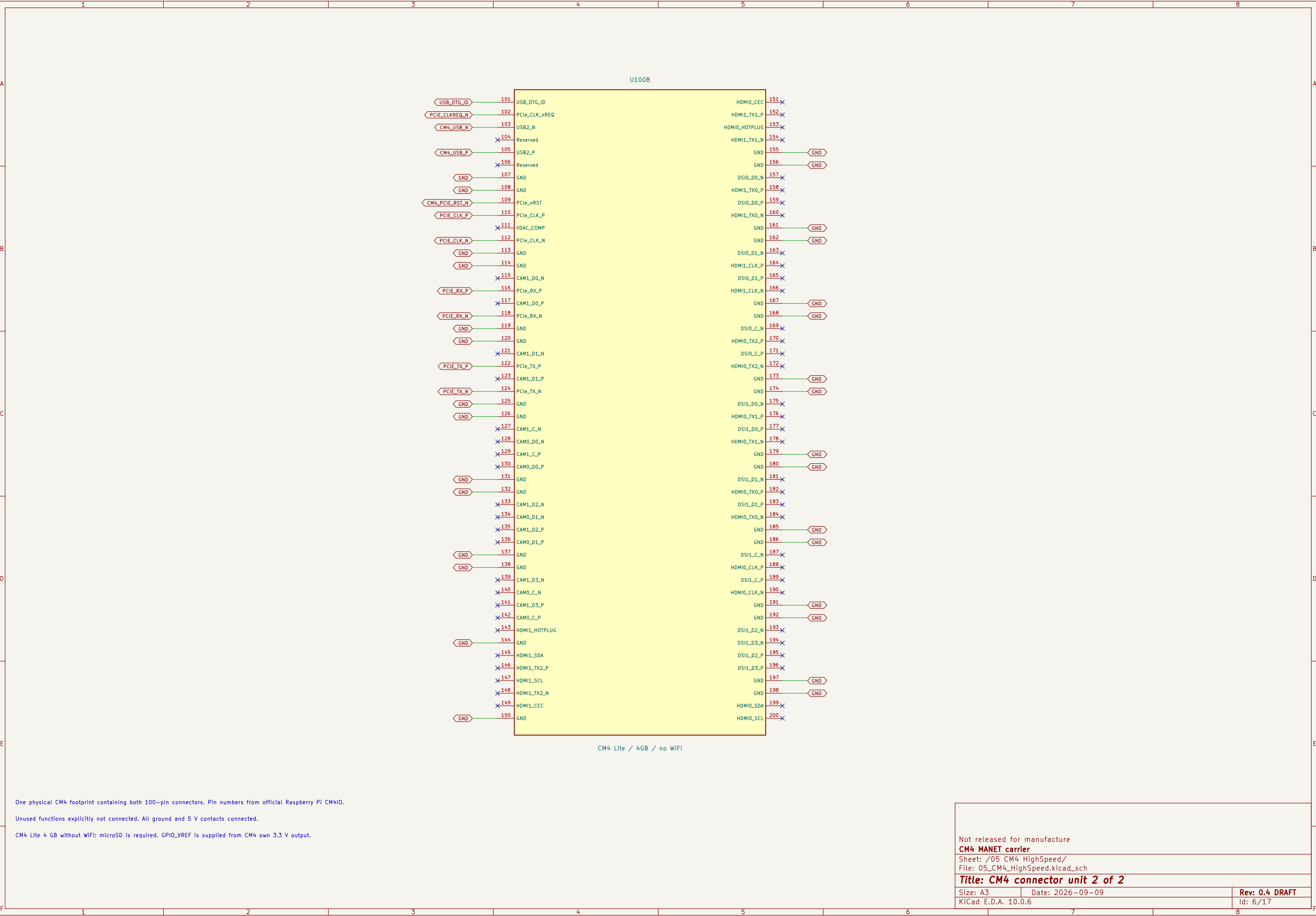
FSW=GND selects nominal 2.5 MHz; DEF=GND selects nominal 3.3 V. FB grounded for fixed-output device.

SS/TR=10nF; nominal soft start about 5ms. Power-good joins open-drain hub reset interlock.

Coil/capacitor effective values and cumulative HaLow decoupling need transient validation.

Not released for manufacture		
CM4 MANET carrier		
Sheet: /03 Aux Power/ File: 03_Aux_Power.kicad_sch		
Title: 3.3 V auxiliary supply for hub and HaLow		
Size: A3	Date: 2026-09-09	Rev: 0.4 DRAFT
KiCad E.D.A. 10.0.6		Id: 4/17





One physical CM4 footprint containing both 100-pin connectors. Pin numbers from official Raspberry Pi CM4IO.

Unused functions explicitly not connected. All ground and 5 V contacts connected.

CM4 Lite 4 GB without WiFi; microSD is required. GPIO_VREF is supplied from CM4 own 3.3 V output.

Not released for manufacture

CM4 MANET carrier

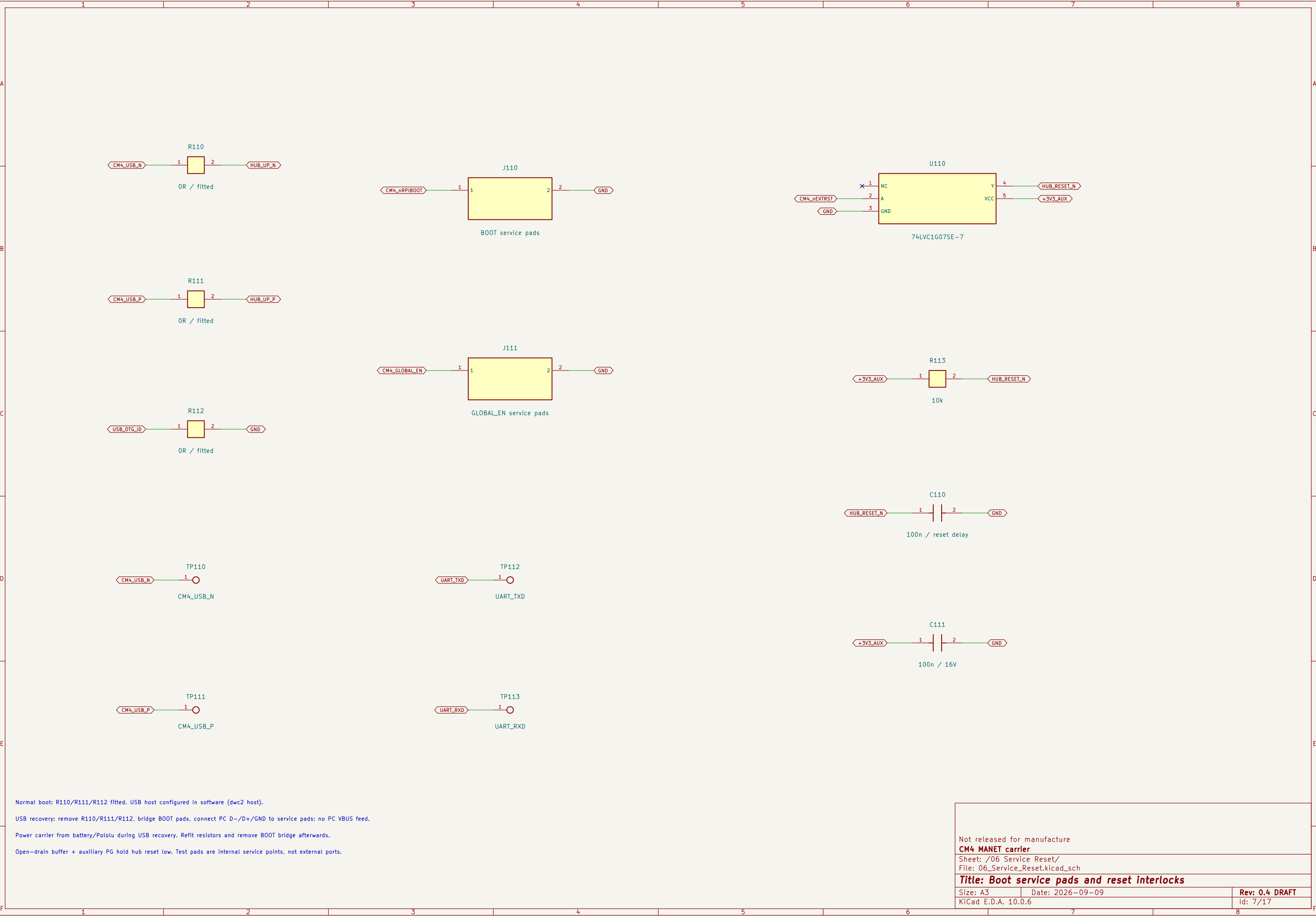
Sheet: /05 CM4 HighSpeed/
File: 05_CM4_HighSpeed.kicad_sch

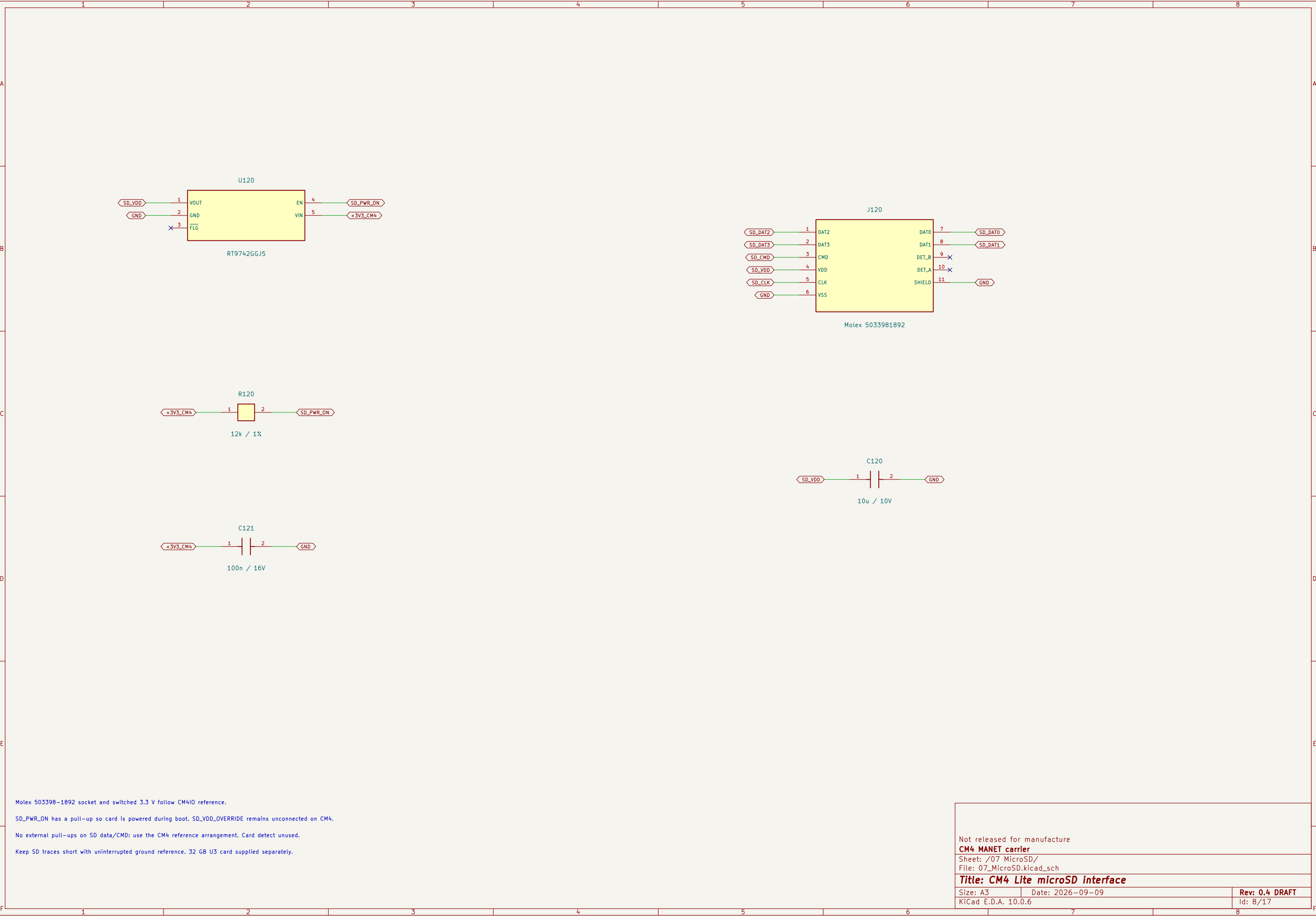
Title: CM4 connector unit 2 of 2

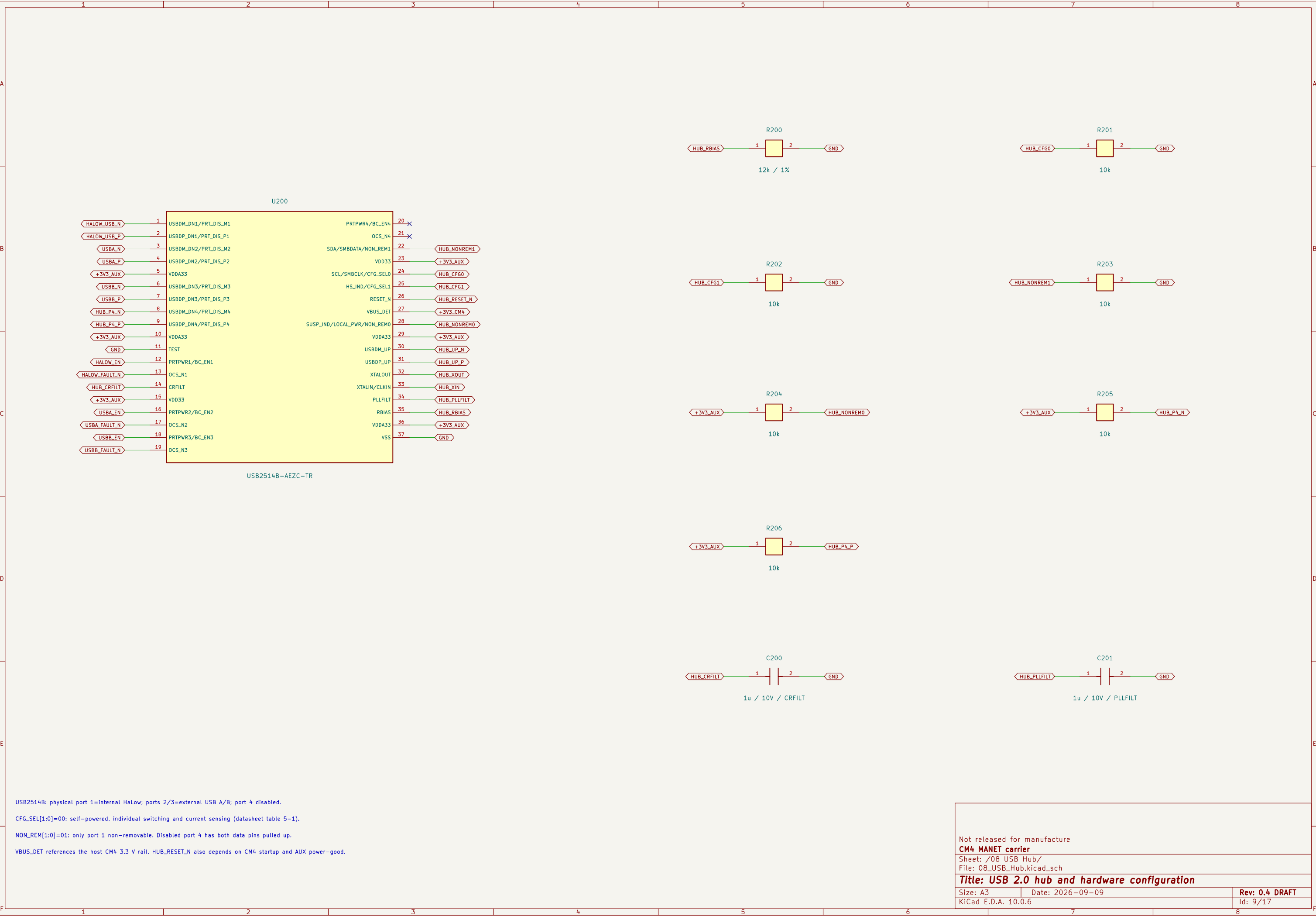
Size: A3
KiCad E.D.A. 10.0.6

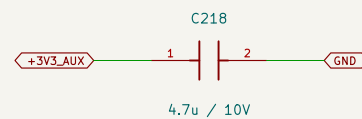
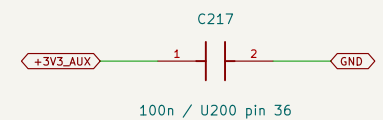
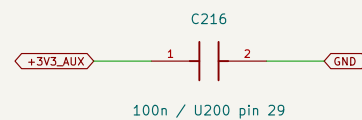
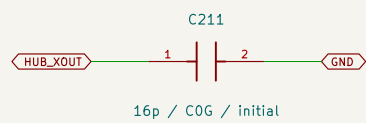
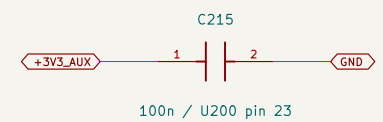
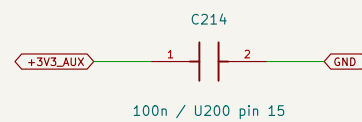
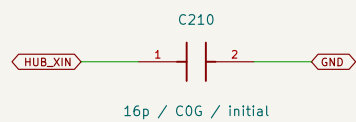
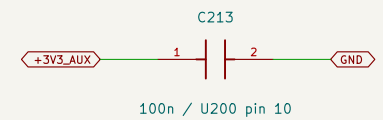
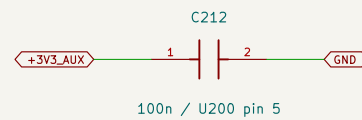
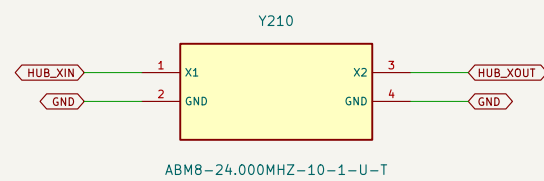
Date: 2026-09-09

Rev: 0.4 DRAFT
Id: 6/17









24 MHz crystal: ABM8-24.000MHZ-10-1-U-T, CL=10pF per Abracon test report.

Initial load capacitors 16pF each assume 2pF equivalent stray. Confirm frequency across tolerance and PCB parasitics.

Each hub supply pin gets its own 100nF placed locally. 4.7uF is the local bulk capacitor.

PLLFLT and CRFLT capacitors are on the hub sheet. Never connect those nodes to an external supply.

Not released for manufacture

CM4 MANET carrier

Sheet: /09 USB Clock Decoupling/
File: 09_USB_Clock_Decoupling.kicad_sch

Title: Hub crystal and local decoupling

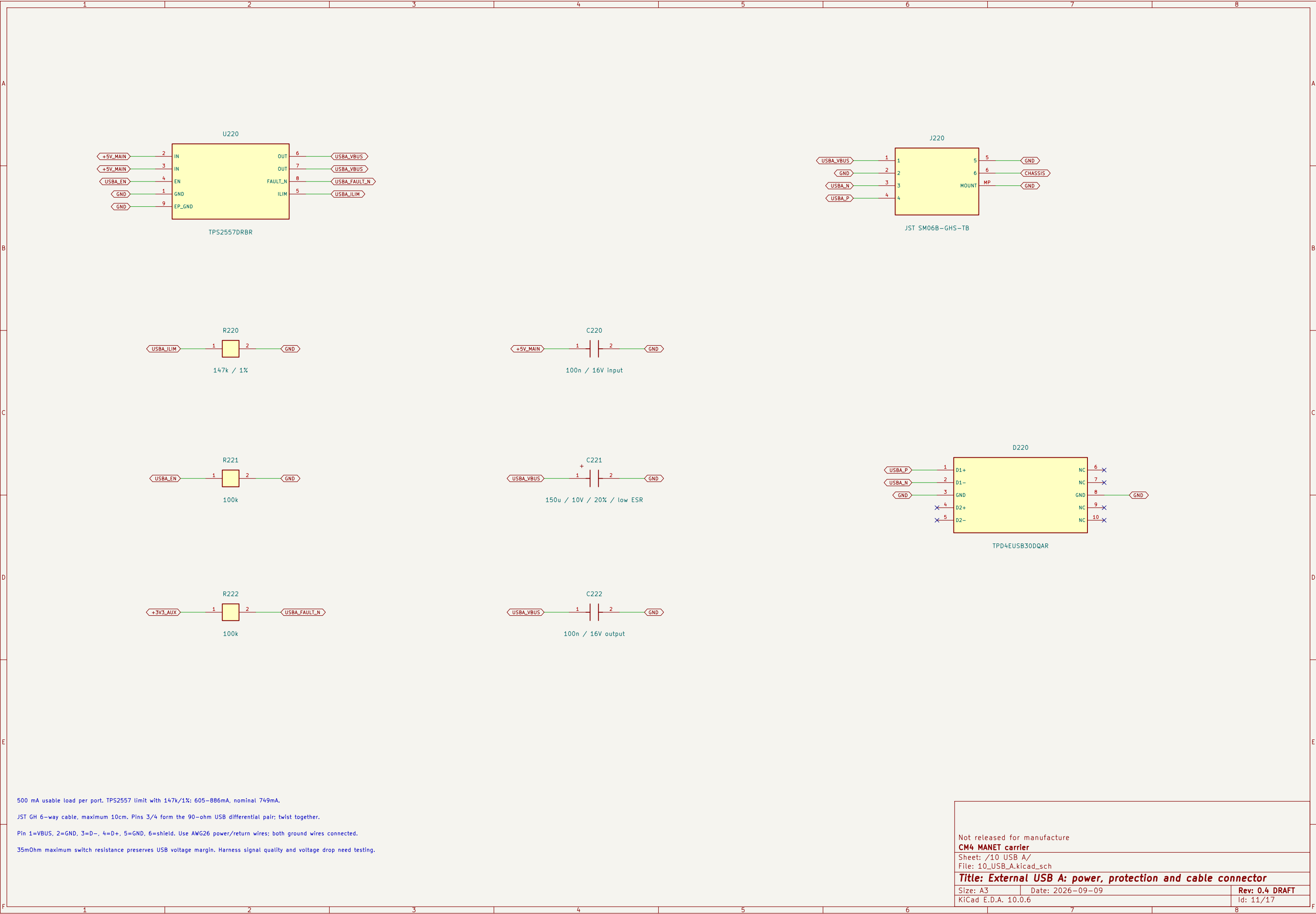
Size: A3

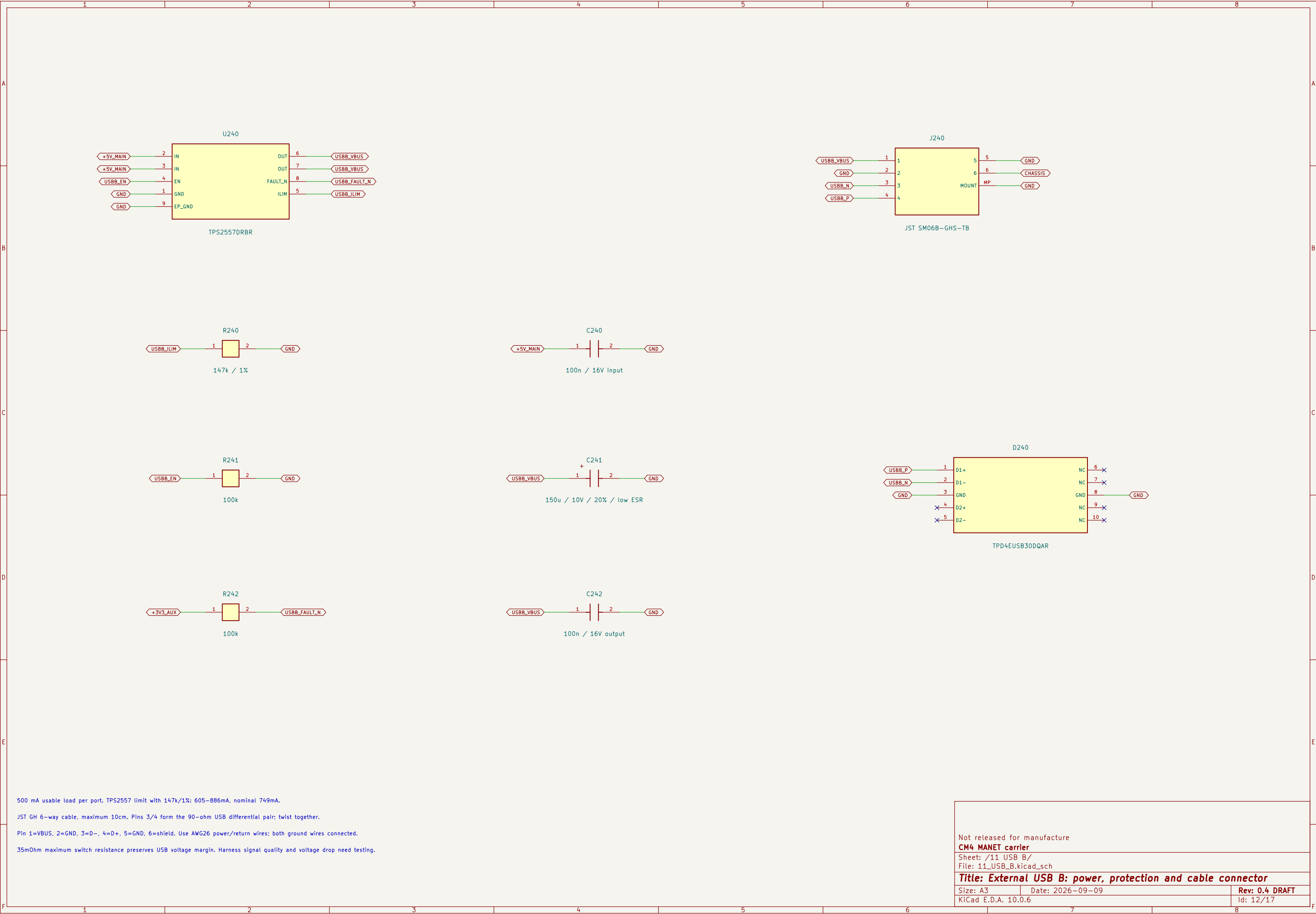
Date: 2026-09-09

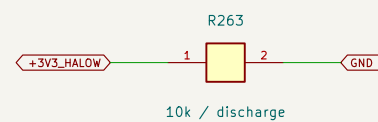
Rev: 0.4 DRAFT

KiCad E.D.A. 10.0.6

Id: 10/17



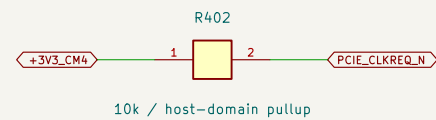
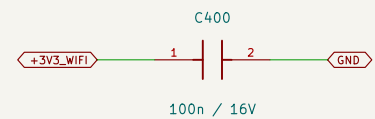
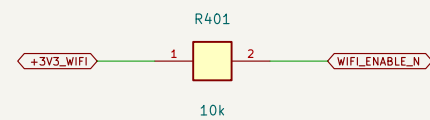
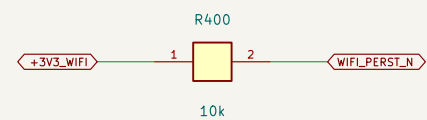
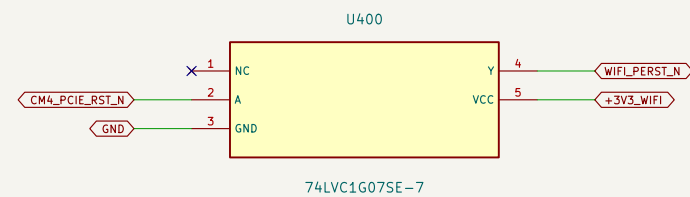
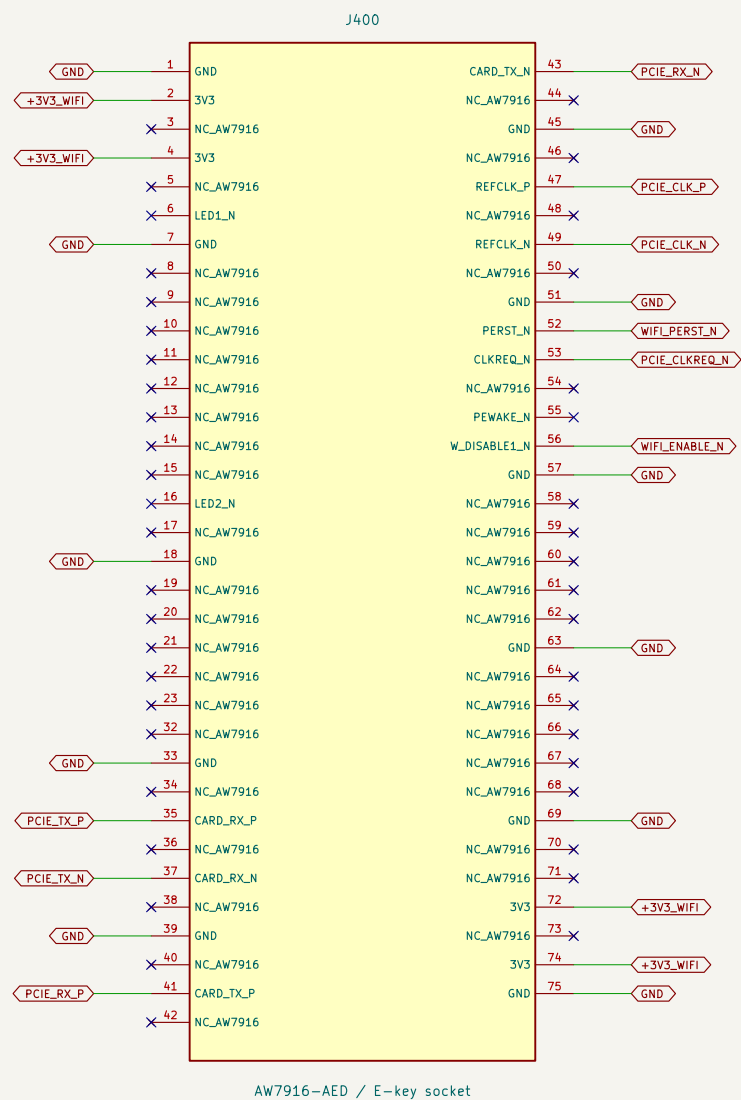




RESET RC is on HaLow sheet. Check turn-off discharge/re-enumeration and 3.0 V minimum at module under peak load.

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AsiaRF drawing names TX/RX from card perspective: card RX pins35/37 receive CM4 TX; card TX41/43 drive CM4 RX.

CM4 already includes PCIe TX and REFCLK coupling. Confirm coupling on AW7916 TX before PCB release.

All four power contacts (2,4,72,74) carry WiFi supply; connector must support at least 0.75 A per power contact.

Exact 1A/contact E-key socket suffix, physical mating and footprint are release blockers. Generic 0.5A sockets rejected.

Not released for manufacture

CM4 MANET carrier

Sheet: /14 WiFi PCIe/

File: 14_WiFi_PCl.e.kicad_sch

Title: AW7916 M.2 A/E PCIe interface

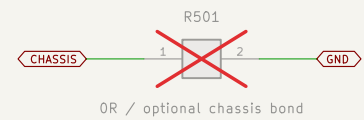
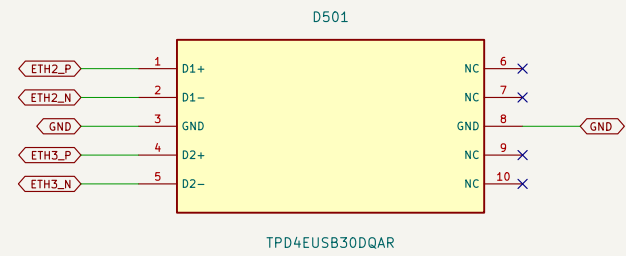
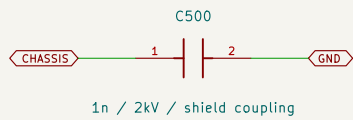
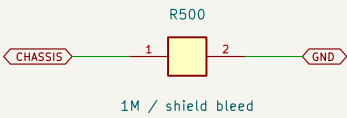
Size: A3

Date: 2026-09-09

Rev: 0.4 DRAFT

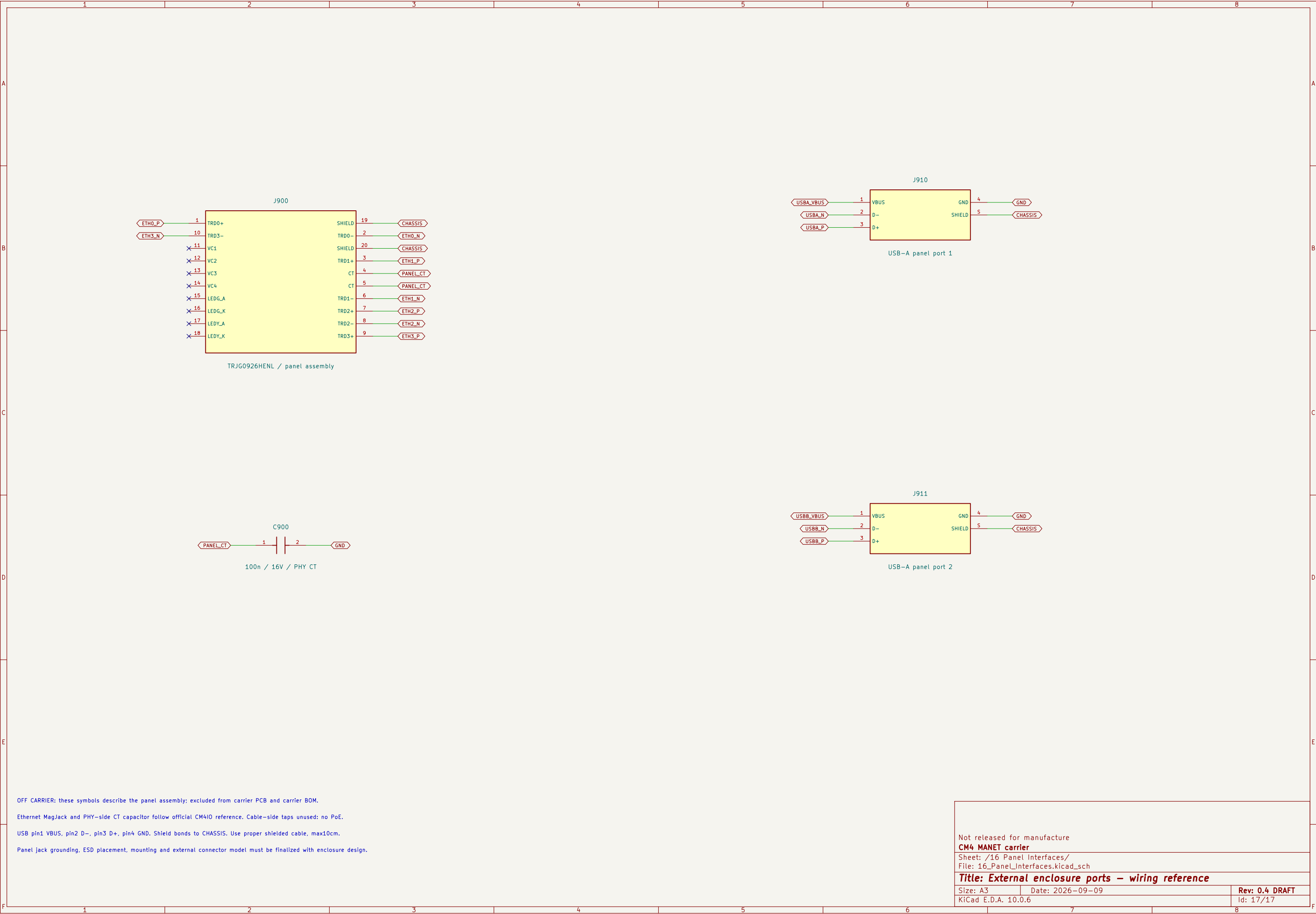
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Id: 15/17



No Ethernet transformer on carrier. External assembly must contain the specified magnetics; plain RJ45 alone will not work.

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J910

1

USBA_VBUS

2

USBA_N

3

USBA_P

4

VBUS

5

D-

6

D+

7

GND

8

SHIELD

9

GND

10

CHASSIS

USB-A panel port 1

J911

1

USBB_VBUS

2

USBB_N

3

USBB_P

4

VBUS

5

D-

6

D+

7

GND

8

SHIELD

9

GND

10

CHASSIS

USB-A panel port 2

C900

1

PANEL_CT

2

GND

100n / 16V / PHY CT

OFF CARRIER: these symbols describe the panel assembly; excluded from carrier PCB and carrier BOM.

Ethernet MagJack and PHY-side CT capacitor follow official CM4IO reference. Cable-side taps unused; no PoE.

USB pin1 VBUS, pin2 D-, pin3 D+, pin4 GND. Shield bonds to CHASSIS. Use proper shielded cable, max10cm.

Panel jack grounding. ESD placement, mounting and external connector model must be finalized with enclosure design.

Not released for manufacture

CM4 MANET carrier

Sheet: /16 Panel Interfaces/
File: 16_PanelInterfaces.kicad_sch

Title: External enclosure ports – wiring reference

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-09

Rev: 0.4 DRAFT
Id: 17/17